CHAPTER TWO

A secure base script perspective on attachment: progress, promise, and prospects

Theodore E.A. Waters[a,*], **Victoria L. Zhu**[b], **Glenn I. Roisman**[b]
[a] New York University – Abu Dhabi, United Arab Emirates
[b] University of Minnesota, Minneapolis, MN, USA
*Corresponding author. e-mail address: theo.waters@nyu.edu.

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Abstract

Bowlby's attachment theory has long been a dominant framework for understanding the development of intimate relationships, particularly the role of early caregiving in shaping later socio-emotional functioning. While Bowlby's proposed cognitive mechanism allowing for early experience to guide later adaptation, the Internal Working Model, has been central to the theory, the specific psychological constructs and processes at play remain underexplored. Waters and Waters (2006) advanced our understanding of the Internal Working Model by introducing the "secure base script," a cognitive representation that summarizes secure base experiences and guides behavior in attachment-related contexts. Twenty years on, this review summarizes current progress in terms of: a) assessment; b) latent structure; c) antecedents; d) sequalae; and e) stability/change. Further, it evaluates the secure base script's potential to advance attachment theory and our understanding of attachment processes by

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integrating both basic and clinically oriented perspectives from cognitive psychology with attachment theory.

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1. Introduction

Bowlby's (e.g., 1969/1982) attachment theory has served as a dominant perspective for understanding the development of intimate relationships across the lifespan for more than a half century. Attachment theory not only serves as a framework for understanding the development of parent-child relationships but also how those relationships inform both our later attachments and our socio-emotional adjustment into the years of maturity (Roisman & Fraley, 2012; Waters & Waters, 2024). For Bowlby, the critical dynamic between parent and child was the provision and use of secure base support. The infant/child's use of their primary attachment figures as a source of support during distress, as well as a secure base from which to explore, is thought to organize the attachment system and build competency at utilizing this secure base relationship.

Those who receive consistent and supportive care during their continuous cycles of exploration and proximity seeking form what Bowlby termed a "secure" attachment, marked by clear signals of distress and need for proximity followed by the acceptance of comfort and an expeditious return to exploration (Bowlby, 1969). However, not all caregivers provide the type of consistent, responsive, and sensitive care required to build a secure attachment. In cases where care is aloof, inconsistent, or even harsh and rejecting the attachment that forms is "insecure" (or as Ainsworth et al., 1978, termed it, "anxious") and marked by tentative or limited proximity seeking, difficulty accepting or rejection of comfort, and a protracted period of distress. Once established, these individual differences in secure base dynamics are internalized as an Internal Working Model (IWM) of attachment and carried forward across development, serving as a guide for behavior and expectations in intimate relationships (Bowlby, 1988). Thus, for attachment scholars, the IWM serves as a primary mechanism by which early caregiving experiences come to exert enduring effects across the lifespan.

Over the decades, attachment researchers have applied rigorous observational, longitudinal, and experimental methods to test Bowlby's predictions. The data largely supports the assumptions contained within attachment theory regarding to origins of attachment being in the early caregiving environment (e.g., Madigan et al., 2024), attachment being carried forward as a (modestly) stable individual difference across development (e.g., Pinquart et al., 2013; Waters et al., 2021a), and its developmental sequelae (e.g., Dagan et al., 2021; 2024a; Groh et al., 2014). Despite the promising body of research supporting attachment theory, the primary mechanism that gives the theory its explanatory power is comparatively less well understood. Said another way, the specific contents of the IWM or the psychological processes that account for its origins, stability, and sequalae have long been underdeveloped in attachment theory and a point of reliable criticism (e.g., Hinde, 1982; 1991). The IWM lacked a degree of specificity due, in part, to the limitations of the cognitive psychology of Bowlby's day which was still developing its theories of mental representations of experience.

Building from Bretherton's seminal work on the IWM (e.g., Bretherton, 1990; see also Main et al., 1985) and the cognitive and developmental literatures on event representations, Waters and Waters (2006) argued that the attachment IWM contains (among other things) a cognitive script summarizing the history of secure base use and support, termed the *secure base script*. This theorizing represented an important step toward specificity and created a meaningful bridge between attachment theory and the extensive literature on event representations (e.g., Nelson, 1986; Schank & Abelson, 1977). Now, nearly twenty years since Waters and Waters' (2006) proposal, we provide a review of the literature on the secure base script. We focus on three key assumptions underlying both attachment and cognitive scripts, mainly that they are (a) tolerably veridical with experiences in the parent-child relationship, (b) a heuristic or guide for behavior/adaptation, and (c) relatively stable once established. Finally, we evaluate the literature on the secure base script in light of the initial promise of this construct for advancing attachment theory and offer several prospects for moving the field forward.

2. Assessment

Waters and Waters (2006; see also Waters et al., 1998) defined the secure base script as consisting of eight elements of secure base support linked in a temporal-causal sequence. From the secure base script perspective, individual differences in attachment security and the IWM are operationalized as varying degrees of knowledge and elaboration of the script. Assessment of the secure base script has generally taken two parallel approaches: 1) fictional storytelling tasks and 2) autobiographical narrative assessments.

2.1 Prompt-word methods

The attachment script assessment (ASA; Waters & Waters, 2021) is the most widely used assessment tool for secure base script knowledge. Versions of the task cover a broad range of development (i.e., middle childhood, adolescence, emerging adulthood, and adulthood) and cultural contexts, including versions in English, Spanish, Dutch, Portuguese, Chinese, and Japanese, with varying degrees of validity data (Waters & Roisman, 2019). Common to all is the utilization of the prompt-word methodology (Waters & Hou, 1987) to elicit fictional attachment narratives which are then rated on a scale from 1 to 7 for the extent to which the narrative follows the secure base script and provides elaboration/detail around the various script elements (e.g., offering explicit detail about comfort, including physical touch and verbal encouragement). The prompt-word method provides participants with a list of words organized in columns that form a beginning, middle, end structure with the instruction being for the participant to tell a story using the outline as a guide and to attempt to fill in as much detail around the outline as they can. Each story in the ASA focuses on a distress scenario in which it would be appropriate to seek support from a primary attachment figure such as a child seeking support after falling off a bike, an adolescent seeking support from a parent after a conflict with peers, or a romantic partner seeking support from a spouse during a health scare. Individual differences in secure base script knowledge are then assessed from transcripts of each story and the average score across a set of three to four attachment narratives is taken to maximize the reliability and validity of the assessment (Waters & Waters, 2021). Neutral/control stories about routine activities with peers are often include for practice or to limit hypothesis guessing.

2.2 Story-stems

The prompt-word method's reliance on literacy and understanding of narrative structure makes it challenging to use with children under the age eight or nine. As such, researchers interested in studying the secure base script in young children have turned to the story stem procedure (e.g., Bretherton & Oppenheim, 2003; Oppenheim & Waters, 1995) to help scaffold children's emerging narrative skills and examine individual differences in secure base script knowledge (Ruiz et al., 2019; Vaughn et al., 2019; Waters et al., 1998). The story stem procedure provides children with various attachment/relationship-relevant scenarios using a set of dolls or toys. The first part of the story is provided by the experimenter and the child is then asked to show and tell what will happen next. These narratives are then coded for the extent to which they reflect the secure base script and averaged across multiple story stems, once again in the interest of maximizing reliability and validity of the overall assessment. Again, this procedure for assessing secure base scripts has been applied in a variety of cultural contexts (see Vaughn et al., 2019).

2.3 Autobiographical interview approaches

Attachment research has a long and well validated tradition of assessing attachment security through autobiographical interview which started with the introduction of the Adult Attachment Interview (AAI; Main et al., 1985). To date, more than 26,000 AAIs (Bakermans-Kranenburg, Dagan, Cárcamo and van IJzendoorn, 2024) have been collected and parallel interviews focused on other developmental periods or relationships (romantic partnership; Current Relationship Interview (CRI); Crowell & Owens, 1996) have been developed as well. Following the initial validation and success of the fictional narrative approaches to assessing secure base script knowledge, researchers began applying similar frameworks for evaluating script knowledge reflected in the autobiographical narratives produced during the AAI (AAI_{sbs}; Waters & Facompré, 2021) and the CRI (CRI_{sbs}; Nivison, et al., 2022). Unlike the traditional coding systems for these interviews that focus on narrative coherence across the entire interview, the secure base script coding focuses on the first portion of the interview that emphasizes more generalized relationship expectations and prompts for specific narrative examples of these expectations. Despite significant differences in the tasks, convergent validity between the ASA and the AAI suggests that both approaches assess the same underlying construct (Waters et al., 2021b).

3. Latent structure

Issues of latent structure are of both theoretical and methodological import. Traditionally, attachment research has approached the assessment of individual differences under two key assumptions that: a) attachment is generalized in that it reflects a quality of the individual rather than that of a specific relationship and b) individual differences should be thought of as differences in kind rather than of degree (e.g., Fraley & Speaker, 2003). The majority of the evidence on the latent structure of attachment indicates that it becomes generalized after infancy and that, in contrast to expectations, individual differences are continuously distributed (e.g., Fraley et al., 2013). The emergence of the secure base script construct naturally renewed interest in these questions, especially in the context of cognitive script theory, which shares similar assumptions about generalization and all-or-nothing individual differences (Waters et al., 2015).

The ASA lends itself well to examining the generalization of secure base script knowledge because both the adolescent and adult versions include separate items for mother/father and parent/romantic partner scenarios, respectively. As such, confirmatory factor analysis can be used to examine the relative fit of a generalization model (covariance between latent variables for mother/father or parent/romantic partner ASA scores is freely estimated) versus a discrete representations model (covariance between latent variables is fixed to zero). Waters et al. (2015) compared the fit of these two models using both the adolescent and the adult version of the ASA. In both cases, evidence strongly supported the generalization model. Waters et al. (n.d.) performed a downward and cross-cultural extension of this work in China. They examined the same models using a new adaptation of the middle childhood ASA in a sample of 10-year-old Chinese children included two mother-child and two father-child items. Again, results supported the generalization model.

Building on work by Fraley and colleagues (e.g., Fraley et al., 2014; Roisman et al., 2007) which utilized taxometric analysis (e.g., Meehl, 1992; 1995; Ruscio et al., 2007) to distinguish between continuous and categorical latent structure, Waters et al. (2015) also tested for categorical individual differences in the same adolescent and adult samples which supported the generalization model. They found that, in contrast to assumptions often made by attachment theorists, secure base script knowledge was best thought of as continuously distributed. Waters et al. (2019a) followed up this work with a downward extension into middle childhood in a sample of 10-year-olds from Western Europe. In contrast to what was observed at later ages, they found that individual differences in secure base script knowledge in childhood were best understood as categorically (i.e., secure v. insecure) rather than continuously distributed. This led the authors, informed by both cognitive and developmental theories on event representations (e.g., Fivush, 1984; Hudson, 1992; Schank & Abelson, 1977), to propose that secure base script knowledge emerges in an all-or-nothing fashion. The later expansion of their social world and the need for secure base support in more diverse contexts during the transition from middle childhood to early adolescence leads to a process of elaboration, nuance, and refinement that results in continuously distributed individual differences.

To test this hypothesis, Houbrechts et al. (2024) conducted a large-scale longitudinal study of the latent structure of secure base script knowledge across the middle childhood period. Specifically, they collected the middle childhood version of the ASA in three separate waves when the participants were 10 years, 11 years, and 12 years old. Replicating Waters et al. (2019a), they found categorical latent structure at age 10 years. At age 11 years, however, the taxometric analyses were inconclusive. By age 12 years, a continuous latent structure had emerged and evidence for the developmental shift in the latent structure of secure base script knowledge was observed. Taken together, the work on the latent structure of secure base script knowledge suggests that across development, we construct a generalized script-like mental representation of our caregiving experiences that is initially learned in an all-or-nothing fashion but shifts toward individual differences in terms of degree (more or less secure) and away from differences in kind (secure v. insecure).

4. Antecedents

Data over the past 50 years support the predicted association between early caregiving quality and the development of attachment (Madigan et al., 2024). The link between any measure of attachment and its theoretical antecedents is a cornerstone of validation. This is especially critical when operationalizing attachment as a cognitive script, as cognitive script theory *also* clearly specifies that scripts are generalized representations based on lived experiences (Schank & Abelson, 1977). As such, any new assessment of script-like attachment representations must replicate this general association between early caregiving and later attachment.

Researchers have produced substantial evidence for the association between higher levels of secure base script knowledge in childhood, adolescence, and adulthood and early maternal and paternal sensitive caregiving as well as abuse and neglect. Steele et al. (2014) reported on the association between the adolescent version of the ASA at age 18 years and maternal (9 assessments) and paternal sensitivity (5 assessments) from infancy through age 15 years in the normative-risk Study of Early Child Care and Youth Development (SECCYD). They found that both maternal and paternal sensitivity positively predicted ASA scores in the target adolescents (see also Vaughn et al., 2016). In addition to ASAs (n = 673), the

SECCYD also collected AAIs during the same age 18 years assessment wave. These AAIs (N = 857) were later coded using the AAI_{sbs} system and Nivison et al. (2024) replicated the findings from Steele et al. (2014) in the SECCYD, with both maternal and paternal sensitivity positively correlating with AAI_{sbs}, although the findings for maternal sensitivity were more robust to the inclusion of covariates than were those observed with paternal sensitivity. Looking at secure base scripts in middle childhood, Runze et al. (2024) found that parental sensitivity at roughly age 6 years was positively associated with secure base script scores three years later.

To extend the work on the developmental antecedents of secure base script knowledge to higher risk contexts, Waters et al. (2017) recoded the existing AAIs collected with the Minnesota Longitudinal Study of Risk and Adaptation (MLSRA) cohort. AAIs were collected at both age 19 years and 26 years and observations of maternal sensitivity were collected across seven waves through the first 13 years of life. A composite of the maternal sensitivity assessments was significantly and positively associated with AAI_{sbs} at both ages. Schoenmaker et al. (2015) found similar results using the ASA at age 23 years in a sample of adoptees, thus accounting for genetic confounds, with maternal sensitivity in early and middle childhood significantly positively predicting script knowledge in young adulthood. In the same MLSRA sample as Waters et al. (2017), Eller et al. (2022) examined the role of consistency in sensitive caregiving in addition to mean levels, as both cognitive script theory and attachment theory also make arguments that the construction and organization of mental representations benefit from consistency across experiences. Supporting this view, they found that individuals who experienced greater fluctuations in maternal sensitivity during childhood showed significantly lower levels of secure base script knowledge in adulthood, even after accounting for mean levels of sensitivity. Finally, Waters et al. (2025) tested for cross-cultural replication of the sensitivity to secure base script knowledge association in a prospective longitudinal study of Chinese families. They found that maternal sensitivity observed at 14 and 24 months positively predicted secure base script knowledge in children at age 10 years. Taken together, the literature demonstrates a significant association between the development of the secure base script and the level and consistency of sensitive caregiving experienced during development in normative risk, high-risk, genetically unrelated, and cross-cultural samples.

Nivison et al. (2021) looked beyond sensitive caregiving as a positive predictor and examined the role of abuse and neglect in secure base script development. They explored the role of perpetrator, timing, and type of abuse in the MLSRA cohort. They found that total abuse/neglect experienced was negatively associated with later secure base script knowledge even after accounting for maternal sensitivity. In terms of timing and type, the effects were present for abuse/neglect experienced during middle childhood and adolescence and specifically for physical abuse. They also found that these effects were unique to maternal and paternal abuse/neglect and did not extend to non-caregiver relationships. The parental caregiver finding parallels work by Waters et al. (2021c) with the SECCYD which found that early child care/daycare experiences (quality, quantity, and type) were unrelated to later secure base script scores. These results provide some discriminant validity to the secure base approach to studying attachment representations in that the effects seem confined to primary attachment figures.

Given the strong emphasis placed on the caregiving environment by attachment theory it is not surprising that the vast majority of research on the origins of individual differences in attachment (secure base script knowledge included) has focused on environmental influences within biologically-related families. As a result, the potentially confounding role of genetics remains largely unaccounted for (but see Schoenmaker et al., 2015). The only study to date looking at the relative contributions of genetic and environmental factors to secure base script knowledge found that genetic influences accounted for 38 percent of the variation in secure script scores in middle childhood (Runze et al., 2024). This result stands in contrast to the sensitivity hypothesis and prior genetically-informed attachment work in infancy, which found minimal evidence for genetic influences on attachment (Bakermans-Kranenburg & Van IJzendoorn, 2016; Bokhorst et al., 2003; Fearon et al., 2006; Roisman & Fraley, 2008). This result, however, was consistent with those reported by Fearon et al. (2014) in sample of adolescents using the traditional coding approach with the Childhood Attachment Interview. Though the genetically informed literature on secure base representations is scant, initial evidence suggests the presence of potential gene-environment correlations. As children develop, their increased ability to engage with environments that align with their genetic predispositions could increase the heritability of secure base script knowledge. As increasing evidence suggests attachment, and the secure base script, is subject to substantial influence from genetics, more research is needed to replicate these initial findings. It will be especially critical to leverage high-powered genetically-informed samples in testing the origins of secure base representations where possible, to control for genetic confounding as well as to look at the genetic and environmental contributions to secure base script knowledge at various stages across the lifespan, particularly given that behavioral geneticists suggest that the heritability of a trait increases over time (Turkheimer, 2000).

5. Sequalae

Over the past two decades, researchers have reported associations between secure base script knowledge and a myriad of theory consistent sequelae including caregiving behavior, social competence, and psychological adjustment. Bowlby suggested that a child who is "much-loved" and sensitively cared for develops a sense of self-confidence that others will see them as worthy of love also; whereas when a child is treated neglectfully or as unwanted by his caregivers, the child comes to internalize this feeling to a more general level (Bowlby 1973, p. 204). As such, a strong secure base script allows an individual to form not only a set of expectations regarding attachment figures and their support but also an adaptive understanding of oneself. In doing so, this affords a sense of confidence in exploring the environment as well as dealing with potentially distressing situations. Conversely, individuals who have low expectations of security and safety from their caregivers, that is, relatively weak secure base script knowledge or an internal working model that is antithetical to the secure base script, may have difficulties regulating their emotions or be at increased risk of maladaptive psychosocial pathways.

5.1 Psychological functioning

Secure base script knowledge can provide a protective effect against problematic psychological development. A secure base script reflects internally-held themes about oneself and one's relationships; when an individual repeatedly experiences harsh or neglectful interactions with attachment figures., their attachment expectations can take on anomalous qualities, such as the child feeling responsible for relieving their caregiver's distress or believing that they are unwanted or flawed. Evidence from middle childhood through late adolescence indicates that higher secure base script knowledge is associated with fewer early maladaptive schemas when concurrently assessed (Li et al., 2024). Research on young adults similarly indicated that higher secure base script knowledge, derived from free recall of close relationships, was inversely associated with four out of five maladaptive schemas related to emotional disconnection with others, such as social isolation and defectiveness (McLean et al., 2014). Thus, the development of script-like representations of early caregiving experiences may serve as a cognitive framework through which children internalize positive and adverse childhood experiences, shaping their psychosocial adjustment.

As children internalize their interactions with caregivers, they may also develop interpersonal skills and an implicit understanding of the self in relation to others that guides their behavior toward or away from psychopathology. Across multiple cultural contexts, researchers have produced converging evidence indicating the negative association between secure base script knowledge and externalizing behaviors (Portugal: Fernandes et al., 2019; Dutch-speaking Belgium: Houbrechts et al., 2023; Waters et al., 2015; Singapore: Cheung et al., 2024).

Longitudinal evidence, from multiple studies utilizing the NICHD SECCYD, indicates that secure base script knowledge is negatively associated with depressive symptoms such that stronger secure base script content at age 18 was associated with fewer depressive symptoms when assessed in adulthood (Dagan et al., 2021; Dagan et al., 2024b). In one such study, secure base script knowledge, assessed via the ASA, partially mediated the relation between parental sensitivity, assessed at multiple time points and developmental stages within the first 15 years of life, and depressive symptoms at age 26. However, it should be noted that the effect size was modest.

Multiple studies highlight the potential indirect role that secure base script knowledge may play in the relationship between environmental stressors and psychological development. Instability within an environment undermines not only a child's ability to successfully regulate emotions and behavior (Ackerman et al., 1999; Bobbitt & Gershoff, 2016), but also their ability to abstract a cognitive script from their early experiences with caregivers. Ruiz and colleagues (2019) found that development of secure base script knowledge in six-year-olds was inversely associated with parent-reported stress exposure, such as death, divorce, and residential moving, and partially mediated the pathway between early life stress and increases externalizing behavior problems at age eight in a high-risk sample. In this study, higher levels of secure base script knowledge predicted smaller increases in externalizing problems relative to children with lower levels of script knowledge. As such, development of the secure base script provides a potential mechanism for explaining the trajectory of psychosocial adjustment, and by this logic, secure base script knowledge may also provide a buffer from psychological symptoms. In a study conducted in Belgium, secure base script knowledge moderated the relationship between daily stress and depressive symptoms in middle childhood (van Aswegen et al., 2023). In addition, script knowledge during middle childhood served as a protective factor/buffer against the detrimental effects of cumulative family stress in relative change of externalizing problems over a two-year period in a normative risk sample (Houbrechts et al., 2023).

The secure base script may buffer responses to stress as it allows an individual to effectively utilize their attachment figures in the face of challenging environments. However, the secure base expectations that are imbued in an individual with a strong secure base script may reveal a vulnerability towards less than favorable conditions later in life. In a sample of primiparous mothers, higher secure base script knowledge moderated the relation between perceived social support and postpartum psychosocial functioning (Gu et al., 2024). Gu and colleagues found that mothers with higher ASA scores but lower perceived social support during the perinatal period reported higher postpartum maternal maladjustment, a composite of postpartum depression, emotion dysregulation, and internalizing and externalizing behavior metrics. The authors argued that the secure base script reflects a set of expectations and when those expectations are contradicted by the environment (in this case, low social support during the transition to motherhood), it results is psychological distress. As such, scripted representations of attachment serve as a set of expectations by which we evaluate and adapt to our current and immediate social environments, not as an inoculation against negative outcomes.

5.2 Social competence

Throughout development, the opportunities for novel social environments inevitably increase. Attending school, socializing with peers of a similar age, and later, exploring romantic relationships all present opportunities to engage with new social partners. As suggested by Bowlby, attachment representations can be generalized to social interactions outside of the immediate caregiving environment (1969/1982, 1973). The internal working model of attachment allows individuals to organize their expectations and behavior within such novel social contexts in accordance with their secure base script, and as such, secure base script knowledge is activated during such interactions and serves to guide or shape how the individual processes and behaves in unexplored social environments (Bowlby, 1973).

5.2.1 With peers

As children shift from an early caregiving environment into a school environment, the dramatic shifts in context allow for ripe opportunities to utilize their script-like cognitive representations to interact with peers and teachers. Secure base script knowledge has been empirically associated with multiple indices of peer and school-based social competencies across multiple cultural contexts. Fernandes et al. (2019) found significant associations between script-like content elicited from the Attachment Story Completion Task (ASCT) and social competence metrics assessed through direct classroom observation and sociometric interviews in a sample of Portuguese and American four- to six-year-old children. Posada and colleagues (2019) replicated these significant positive associations between secure base script knowledge, assessed via the Attachment Story Completion Task, and teacher reports of social competence in two independent samples of US children drawn from Midwestern and Southern contexts with a moderately strong effect size. Similar associations in preschool-aged children have also been observed in South Korean (Shin, 2019), Portuguese (Fernandes et al., 2019), Mexican, and Peruvian samples (Nóblega et al., 2019).

Researchers have also sought to understand how the secure base script can bolster positive prosocial competencies in children and buffer the effects of negative school-based experiences. Secure base script knowledge, assessed in middle childhood, has been shown to moderate the relation between Reactive Attachment Disorder (RAD) symptoms and parent and teacher-reported prosocial behavior such that in children with more RAD symptoms, those who had higher secure base script scores exhibited higher prosocial behavior (Cuyvers et al., 2020). Similarly, research in Singapore suggests that higher secure base script knowledge is negatively associated with self-reported peer victimization and bullying and positively associated with academic performance (Cheung et al., 2024).

Within an urban Chinese sample, researchers have found that secure base script knowledge in middle childhood partially mediates the relation between early caregiving sensitivity, measured during the first two years, and later friendship conflict at age 15. Yang et al. (2024) found that higher maternal sensitivity was longitudinally associated with lower adolescent conflict within friendships and that ~19 percent of this effect was mediated by secure base script knowledge. Further, the same study found that adolescents with lower secure base script reported greater loneliness in high friend/peer conflict environments . This potentially highlights how the secure base script may guide expectations and interpretations of peer conflict so as to buffer from perceptions of loneliness.

5.2.2 With romantic partners

Attachment representations of early caregiving experiences have been shown to reliably predict variation in multiple indices of romantic functioning (e.g., Dagan et al., 2021; Dagan et al., 2024a; Nivison et al., 2022; Roisman et al., 2001; Waters et al., 2018). Utilizing the SECCYD ample, researchers found that participants' scripted representations of attachment at age 18 years, indexed by the Attachment Script Assessment, mediated the relationship between parental sensitivity assessed in multiple waves across the first 15 years of life and perceptions of romantic partner's extreme hostility at age 26 (Dagan et al., 2021). Further, secure base script content extracted from the AAI at age 18 predicted high romantic dyad adjustment scores at age 30 years (Dagan et al., 2024a). In a high-risk sample, secure base script knowledge in early adulthood was found to mediate the effect of maternal sensitivity quality and consistency on romantic relationship effectiveness at age 32 years such that higher mean levels of early sensitivity predicted higher reports of relationship competency via higher script knowledge (Eller et al., 2022).

5.3 Caregiving outcomes

A script-like representation of attachment and secure-base interactions provides caregivers with both expectations for how secure base interactions *should* occur as well as a cognitive script from which to draw upon when caregivers are faced with attachment scenarios in real time. When a caregiver develops a secure attachment representation (i.e., there is strong evidence of understanding the secure base script), this caregiver is hypothesized to behave in accordance with their understanding of secure base interactions and as such, provide appropriate care in accordance with the secure base script. Additionally, the caregiver's secure base script knowledge would also inform their expectations for the child's actions, affect, and responses, thus contributing to the dyadic quality of secure base interactions. Parenting sensitivity has been widely studied as an associated outcome of attachment representations in both normative and high-risk populations (e.g., Coppola et al., 2006; Raby et al., 2021; Van Ee et al., 2017; Witte et al., 2023). Raby and colleagues found that in parents that were Child Protective Services-referred as well as low-risk parents, higher SBSK was associated with more sensitive caregiving during interactions with middle childhood-aged children

(Raby et al., 2021). In a highly stressful caregiving environment, secure base script knowledge can modulate the types of care that caregivers provide. For example, mothers with a history of childhood maltreatment with higher secure base script knowledge, using both the ASA aggregate score as well as only parent-child stories, were observed to have more positive parenting behaviors with their infants (Huth-Bocks et al., 2014). Evidence from asylum-seeking and refugee families indicates that when caregivers have less access to the secure base script, post-traumatic stress disorder symptoms are strongly associated with an increase in insensitive parenting behaviors (Van Ee et al., 2017). Further, utilizing a genetically-informed twin design, Witte and colleagues found that parent secure base script knowledge was significantly associated with both parental sensitivity and sensitive discipline and that these associations were not associated with child-based genetic factors (2023).

Taken together, this research has established a link between parents' secure base script knowledge and caregiving sensitivity, suggesting that in both normative and high-risk caregiving contexts, parents possessing a well-developed secure base script are more likely to engage in sensitive and responsive parenting. Research now highlights the potential neural mechanisms underlying this association. For example, there is evidence that mothers with lower secure base script knowledge exhibit significantly elevated P3b event-related potential (ERP) responses when attending to infant distress, as well as a disproportionate allocation of cognitive resources toward distress cues from their infants rather than positive cues. This pattern suggests that low SBSK mothers engage more cognitive resources in processing attachment-related signals compared to mothers with higher SBSK as well as have a bias in their attachment-signal related processing (Groh & Haydon, 2017). Further, there is early evidence that maternal secure base script knowledge may affect neural development in the next generation, particularly in child brain structure (Fitter et al., 2022).

Researchers have established strong support for the intergenerational transmission of attachment security (e.g., Van IJzendoorn, 1995; Vaughn et al., 2007; Verhage et al., 2016). Specifically for scripted attachment representations, research indicates that parental secure base script knowledge may predict child attachment security (Waters et al., 2018). Similarly, Huth-Bocks et al. (2022) found a longitudinal association between maternal SBS knowledge and child attachment behavior during infancy in a sample of caregivers at risk of parenting difficulties. One potential explanation in this intergenerational attachment representation transmission may be that the secure base script may play a role in how mothers jointly discuss emotional content with their children (Bost et al., 2007). Although secure base script knowledge has been shown to increase parenting sensitivity, there is also evidence that a strong command of the secure base script increases parents' likelihood and encouragement to openly engage with their children in emotion-based speech (Apetroaia & Waters, 2018; Waters et al., 2018). Relatedly, in understanding potential mechanisms underlying how secure base script knowledge affects parenting, research has also indicated indirect associations between secure base script knowledge, self-reported attachment styles in parents and their reflective functioning (Borelli et al., 2017). In this cross-sectional study, Borelli and colleagues found that lower secure base script knowledge was associated with higher self-reported attachment avoidance in parents as well as less positive/more negative emotionality after reflecting on positive connections with their child (Borelli et al., 2017).

Parent secure base script knowledge has also demonstrated long-term predictive validity in intergenerational outcomes for academic success, particularly in at-risk populations. For example, McLear et al. (2016) found that parents' SBSK uniquely predicted their kindergarteners' academic achievement at the end of the school year, above and beyond the effects of effortful control and child verbal ability. This suggests that, beyond cognitive and self-regulatory skills, a parent's internalized attachment representations contribute significantly to their child's educational trajectory. Similarly, parent ASA scores have been linked to child social competence, an important predictor of academic engagement. Sparks (2018) demonstrated that parental attachment script knowledge (ASA) was positively associated with children's social competence, which in turn mediated the relationship between family cumulative risk and social development.

Beyond caregiving and outcomes in the next generation, research also suggest that higher secure base script knowledge predicts the quality of interactions between adult caregivers and their elderly parents (i.e., less criticism and hostility), when there was a high level of perceived caregiving stress (Chen et al., 2013). This work points not only to how secure base script knowledge influences caregiving across different life stages but also to the cognitive script's role in shaping relational sequelae beyond the parent-child dyad. By providing an abstract framework for navigating attachment-related interactions, SBSK may serve as a protective factor in stressful and emotionally demanding caregiving contexts, such as supporting aging parents (Waters & Waters, 2024).

Though a large body of research has established the predictive validity of secure base script knowledge for a range of psychological and social outcomes, there is growing interest in understanding how attachment representations shape physiological development. Initial empirical evidence is inconclusive given the relatively small number of studies that specify relations between secure base script representations and physical health indices. For example, research from the Minnesota Longitudinal Study of Risk and Adaptation indicates that early caregiving sensitivity during infancy in an at-risk sample predicted an composite index of cardiometabolic health (i.e., mean arterial pressure (MAP), body mass index (BMI), C-reactive protein (CRP), and blood pressure) at ages 37 and 39 with secure base script knowledge in early adulthood (ages 19 and 26) partially mediated this relationship (Farrell et al., 2018). However, research utilizing a normative-risk prospective longitudinal sample did not find evidence of secure base script knowledge mediating the relationship between observed parental sensitivity from infancy to age 18 and BMI at age 26 (Dagan et al., 2021). As we continue to consider the multifinality of attachment representations, there is an increasing need for further research investigating the role of secure base script knowledge on markers of physical well-being and psychophysiological development.

6. Stability and change

Beyond typical methodological concerns related to reliability, stability is a critical assumption of attachment theory. Attachment theory does allow for lawful change but is also assumes a degree of stability which supports the mechanistic role attachment representations play in the association between early caregiving experiences and later attachment relevant sequalae. Cognitive script theory also predicts that there should be some degree of stability in scripts and that, assuming experiences in the script domain continue to share common features, this stability may increase over time (Waters et al., 2021d).

Several studies have examined the stability of secure base script knowledge across moderate and long-term intervals, at various periods of development, and across risk status. Vaughn et al. (2006) found that ASA scores were moderately stable over roughly a one-year interval in a small sample of normative risk mothers. This provided the first piece of evidence that the secure base script demonstrated theory-consistent levels of stability over a significant interval. At the same time, the normative risk nature of the sample along with modest sample size prompted several replications and extensions. Waters et al. (2019b) examined ASA stability across the transition from middle childhood to adolescence, a period where greater instability might be expected given socio-emotional and cognitive changes in the child, as well as a shift in the parent-child relationship dynamic toward a supervisory partnership (Koehn & Kerns, 2015; Waters & Waters, 2024). Similar to Vaughn and colleagues (2006), they found moderate stability in ASA scores from age 11 years to 14 years (4-waves with 1-year intervals) of age in a study of normative risk children (see also Houbrechts et al., 2024). Moderate stability was again observed with the same sample following a final 2-year follow-up at age 16 years (Waters et al., 2021d).

Greater instability in attachment may also be expected in higher-risk contexts due to the greater potential for significant changes to relationship dynamics and increased stress combined with less flexibility/availability of attachment figures due to socioeconomic circumstances (e.g., Pinquart et al., 2013). However, secure base script stability has generally produced equivalent stability data irrespective of context. In the MLSRA, the high-risk longitudinal cohort, AAI_{sbs} scores were stable from age 19 years to age 26 years with a comparable effect size to the findings discussed above. One potential confound in stability estimates provided for the secure base script comes from shared method variance. The use of the same measure across multiple timepoints may artificially inflate stability due to consistency driven by the task itself rather than the underlying construct. To address this concern, Waters et al. (2021) examined the stability of the secure base script using two different assessment tools, the ASA and AAI. In a pooled sample of both high and low-risk cohorts ($N = 113$), they found that the AAI_{sbs} assessed at age 20 years significantly predicted the ASA at age 40 years ($r = 0.43$) to a roughly equivalent degree as in prior tests (medium and longer term) of stability. Further, they found that the association was not moderated by risk status.

Despite the notable stability in the secure base script, plenty of room for change exists and attachment and cognitive script theories make several predictions of what governs such changes. To date, however, very little work has examined changes in secure base script knowledge. One study examined change in the secur ebase script over a three-year period and found that the number of self-reported daily hassles (e.g., conflicts with peers) but not big stressors (e.g., parental divorce; death of a loved one) predicted *increases* in SBSK during middle childhood and early adolescence (Waters et al., 2019b). The authors argue that frequent opportunities to practice secure base use with manageable stressors allow for the reinforcement and elaboration of the secure base script. Big events, which are more likely to cause large shifts in family dynamics or be challenging to resolve through secure base support, were unrelated to changes in script scores but this may have resulted from low base rates given the normative risk sample examined. In a somewhat higher-risk context, Finet et al. (2021) found that a sample of internationally adopted children did not differ from their non-adopted comparison group in terms of secure base script knowledge in middle childhood, although they were more likely to have been insecurely attached when first observed a few months after adoption had taken place. Further, concurrent maternal sensitivity was positively associated with children's secure base script scores. Together, these results suggest that with time, supportive care can shift attachment security and promote secure base script knowledge even after significant life changes like international adoption. There is also emerging evidence that secure base script knowledge can be changed through clinical intervention. Raby et al. (2021) evaluated the effectiveness of the Attachment and Biobehavioral Catch-up (ABC) intervention on secure base script knowledge. They examined the impact of ABC on ASA scores in a sample of mothers who had been referred to Child Protective Services due to risk for child maltreatment when their children were infants. Seven years after completing ABC, mothers were tested using the ASA. Mothers who had been randomized to receive the ABC intervention had higher ASA scores than those who had received a control intervention and were equivalent to a sample of low-risk comparative control mothers. Although not specifically designed to target the secure base script, ABC did lead to improvements. Interestingly, one of the main targets of ABC is maternal sensitivity and mediational analyses suggested that ABCs effects on ASA scores was through increasing sensitive caregiving via intervention.

The encouraging theory-consistent stability and change results for the secure base script construct offer numerous opportunities to extend the cognitive script perspective on attachment, especially in the context of studying lawful change. Although the data suggests increasing stability in secure base script knowledge from middle childhood and beyond, early intervention does seem promising. The only intervention work to examine change in SBSK followed up with participants several years after intervention. It is still unclear if the ABC intervention directly alters script knowledge or whether this effect is more an indirect effect resulting from changes in caregiver sensitivity and improvement of parent-relationship quality. Interventions that directly target the secure base script are still in development and represent another promising research avenue as how best to affect change in the secure base script is still very much an open question.

7. Conclusion: Promise and prospect

The initial introduction of the secure base script construct was not met with immediate uptake, as is to be expected in a mature field such as attachment. Any new measure or perspective is going to be confronted with the burden of overcoming the "old wine in a new bottle" problem when entering into a research area with a rich tradition of well-validated assessments across the lifespan (Waters, 2021; Waters & Roisman, 2019). The evidence reviewed above speaks to the validity of the secure base script construct, but less so to the novelty and added value of adopting a cognitive script perspective on attachment. The data support the predictions, stemming from both attachment and cognitive script theory, that the secure base script is learned from early attachment relevant experiences, stable and generalized, and applied to novel attachment relevant contexts.

However critical to validity they are, these findings do not necessarily speak to the promise the cognitive script approach offers. Waters et al. (2013) argued that the standard approach to assessing attachment representations in adolescence and adulthood through the AAI coherence coding was troublingly distant from the constructs thought to give rise to, or result from, attachment representations. Theory and evidence are sparse when it comes to accounting for how early caregiving experiences result in coherent autobiographical narratives in the AAI. It is perhaps even more difficult to account for how coherent autobiographical narratives produce behavior in novel contexts. The rich tradition of cognitive script research does provide a straightforward account for how scripts are acquired, generalized, and guide behavior in novel contexts given their high heuristic value and integration with goal structures (Markman, 1999). However, relying on making inferences from the cognitive literature to account for attachment processes falls short of a true cognitive account of attachment. Attachment researchers need to demonstrate that the secure base script has similar impacts on basic cognitive processes like perception, memory, and decision making as has been seen in the cognitive literature in order implicate specific cognitive mechanisms underlying the associations between the secure base script and various attachment processes.

The clinical promise and potential of the cognitive perspective on attachment has also yet to be realized. The work by Raby et al. (2021) demonstrated that the secure base script may be impacted by intervention efforts, even though the intervention examined was not designed specifically with the secure base script in mind. Instead, the ABC intervention program studied focuses on parental coaching with live feedback targeting sensitive caregiving behaviors (e.g., Dozier & Bernard, 2017). There is, however, a substantial clinical literature focused on intervening with cognitive scripts and schemas which becomes relevant once attachment scholars take a cognitive script approach. Cognitive Behavior Therapy (CBT; Hollon & Beck, 1994; Seligman & Ollendick, 2011; see also Schema Therapy, Young, et al., 2003) is a problem focused approach which attempts to alter both maladaptive behaviors and cognitions through cognitive restructuring and reinforcement and has generally produced modest but consistent positive outcomes (Bosmans, 2016; Wenzel, 2017). However, CBT and related approaches have largely not targeted attachment related clinical concerns, in part because attachment theorists had not fully articulated the cognitive underpinnings of attachment beyond the IWM construct. Bosmans (2016) has argued that the cognitive script perspective has opened the door for integration of some of the well-worn paradigms and approaches from CBT with the parenting and sensitivity focused approaches already being used in the attachment field. The development of new integrative interventions that leverage the strengths of both the cognitive and attachment traditions to better understand and effect change represents a great opportunity for future research.

In short, the cognitive script perspective has made tremendous progress in its first twenty years. This has been facilitated by a highly adaptive set of methods and benefited from the wealth of longitudinal studies available in the attachment field (e.g., MLSRA; SECCYD). Waters and Waters (2006) argued for the benefits of the explanatory power of cognitive scripts for the attachment literature, noting that:

> *"Given their impact on social perception, expectations, and memory, not to mention self-definition and goal structures (e.g., Carver & Scheier, 1990; Epstein, 1994; Horowitz, 1988; Lazarus, 1991), studying script-like representations of secure base experience should have far-ranging and fundamental implications for our understanding of attachment related cognitions, emotions, and behavior. (p. 188)"*

This promise has yet to be fully realized, but with a set of well-validated tools and clear developmental, cognitive, and clinical theory as a guide, the future of the cognitive script perspective on attachment has great potential.

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